

COMPANION SLIDES

Chapter 12

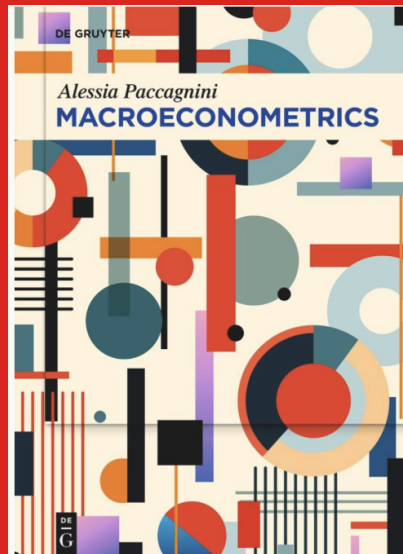
Quantile Regression and Growth-at-Risk

From average outcomes to downside risk

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Means are not enough

How can we measure and forecast the downside risks hidden inside the conditional distribution of macroeconomic outcomes?

- ▶ Standard regression summarizes the conditional mean.
- ▶ Risk management and financial stability require information about tails.
- ▶ Quantile regression lets predictors affect different parts of the distribution differently.

The distribution itself becomes the object of interest.

Foundations

- ▶ Conditional quantiles
- ▶ Check function
- ▶ Estimation and inference
- ▶ Interpreting quantile coefficients
- ▶ OLS vs. quantile regression
- ▶ Quantile VARs

Risk applications

- ▶ Quantile crossing
- ▶ Density forecasting
- ▶ Growth-at-Risk
- ▶ Inflation- and Climate-at-Risk
- ▶ US NFCI application
- ▶ Calibration and risk assessment

- ▶ Explain why conditional quantiles provide information that OLS cannot.
- ▶ Derive and interpret the asymmetric check loss.
- ▶ Estimate and interpret quantile-specific coefficients.
- ▶ Explain QVARs, quantile crossing and rearrangement.
- ▶ Construct direct multi-horizon quantile forecasts.
- ▶ Define and interpret Growth-at-Risk.
- ▶ Evaluate quantile forecasts using coverage and proper scoring rules.
- ▶ Discuss extensions to inflation, financial stability and climate risk.

$$Q_{\tau}(y_t \mid x_t) = \alpha(\tau) + \beta(\tau)x_t, \quad \tau \in (0, 1)$$

- ▶ $\tau = 0.50$: conditional median.
- ▶ $\tau = 0.10$: conditional 10th percentile.
- ▶ $\tau = 0.90$: conditional 90th percentile.
- ▶ Both intercepts and slopes can vary with τ .

One predictor can have very different effects across the distribution.

$$\rho_{\tau}(u) = u(\tau - \mathbf{1}\{u < 0\})$$

- ▶ Quantile regression minimizes asymmetric absolute loss.
- ▶ At $\tau = 0.50$, the loss is symmetric and yields median/LAD regression.
- ▶ At low quantiles, positive and negative residuals receive different penalties.
- ▶ The asymmetry moves the fitted line toward the desired conditional quantile.

$$(\hat{\alpha}(\tau), \hat{\beta}(\tau)) = \arg \min_{\alpha, \beta} \sum_{t=1}^T \rho_{\tau}(y_t - \alpha - \beta x_t)$$

- ▶ The objective is convex and can be solved by linear-programming methods.
- ▶ Asymptotic inference depends on the density of the error at zero.
- ▶ Bootstrap inference is often attractive in finite samples.
- ▶ For time series, block bootstrap methods preserve serial dependence.

Example from the chapter

Suppose tighter financial conditions imply $\beta(0.10) = -1.50$, $\beta(0.50) = -0.60$, and $\beta(0.90) = -0.10$.

- ▶ The predictor has a much stronger association with the left tail than with the median.
- ▶ The right tail is barely affected.
- ▶ This is a statement about the **conditional distribution**, not a regression run only on “bad-growth observations.”

OLS

- ▶ conditional mean
- ▶ squared loss
- ▶ one slope per predictor
- ▶ sensitive to large residuals
- ▶ natural for average effects

Quantile regression

- ▶ conditional quantile
- ▶ check loss
- ▶ slope varies with τ
- ▶ distributional heterogeneity
- ▶ natural for tail risk

- ▶ The research question concerns downside or upside risks rather than averages.
- ▶ Effects may be asymmetric across the outcome distribution.
- ▶ The conditional distribution is skewed or fat-tailed.
- ▶ Extreme observations are economically meaningful rather than mere measurement errors.
- ▶ Policy decisions depend on adverse scenarios and tail probabilities.

Tail estimation needs enough data: extreme quantiles are demanding in short macro samples.

$$Q_{\tau}(y_t \mid \Omega_{t-1}) = A_0(\tau) + A_1(\tau)y_{t-1} + \cdots + A_p(\tau)y_{t-p}$$

- ▶ QVARs extend quantile regression to multivariate dynamic systems.
- ▶ Dynamic relationships can differ across quantiles.
- ▶ Financial conditions may therefore matter more for downside growth than for median or upside growth.

- ▶ With K variables, p lags and Q quantiles, parameter counts grow quickly.
- ▶ The chapter notes that a four-variable, four-lag system at nine quantiles exceeds 2,400 parameters.
- ▶ Possible responses include equation-by-equation estimation, regularization, Bayesian shrinkage and factor reduction.

Distributional flexibility comes with a dimensionality cost.

The problem

Estimated quantiles should satisfy $Q_{\tau_1}(y) \leq Q_{\tau_2}(y)$ whenever $\tau_1 < \tau_2$, but finite-sample estimates may cross.

- ▶ Crossing produces an incoherent predictive distribution.
- ▶ The Chernozhukov–Fernandez-Val–Galichon rearrangement sorts fitted quantiles to restore monotonicity.
- ▶ It is a simple post-estimation correction with sensible fitted values and forecasts.

Iterated

Estimate a one-step QVAR and iterate it forward while maintaining the quantile.

Direct

Estimate a separate quantile regression for each forecast horizon h .

$$Q_{\tau}(y_{t+h} \mid \Omega_t) = \alpha(\tau, h) + \beta(\tau, h)'x_t$$

The GaR framework uses the direct approach; overlapping horizons require HAC-style inference.

- ▶ **Quantile score / pinball loss**: evaluates a forecast at a specific τ .
- ▶ **Coverage**: realized outcomes should fall below the τ forecast approximately τ percent of the time.
- ▶ **CRPS**: summarizes performance across the predictive distribution.
- ▶ Good distributional forecasts should be both calibrated and informative.

Vulnerable growth

Financial conditions predict not only average future growth, but the entire conditional distribution – with particularly strong effects on the left tail.

$$Q_{\tau}(\Delta y_{t,t+h} \mid \Omega_t) = \alpha(\tau, h) + \beta(\tau, h)' X_t$$

Financial vulnerabilities can appear in the tail before they dominate the central forecast.

$$\text{GaR}_{q,t}(h) \equiv Q_q(\Delta y_{t,t+h} \mid \Omega_t)$$

- ▶ $q = 0.05$: severe downside risk.
- ▶ $q = 0.10$: moderate downside risk with more stable tail estimation.
- ▶ A four-quarter horizon is common in policy applications.
- ▶ GaR is analogous in spirit to Value-at-Risk, but applied to future economic growth.

- ▶ Tightening can make credit constraints bind for marginal borrowers.
- ▶ Falling asset prices affect leveraged balance sheets.
- ▶ Higher uncertainty can delay investment and hiring.
- ▶ These channels can raise the probability and severity of contractions without shifting the whole distribution uniformly.

The shape of risk changes, not just the mean.

- ▶ The chapter emphasizes that GaR can deteriorate before recessions materialize.
- ▶ In the literature, downside forecasts worsened materially in the run-up to the Global Financial Crisis.
- ▶ This leading behavior is what makes GaR useful for macro-financial surveillance.
- ▶ Point forecasts can remain relatively benign while the left tail becomes much worse.

Inflation-at-Risk

Quantify risks of unusually high or low future inflation.

Financial Stability

Link current vulnerabilities to future credit, output or systemic tail outcomes.

Climate-at-Risk

Condition growth quantiles on physical or transition-risk indicators.

$$Q_{\tau}(\Delta y_{t,t+h} \mid \Omega_t) = \alpha(\tau, h) + \beta_f(\tau, h)FCl_t + \beta_c(\tau, h)Climate_t$$

- ▶ Physical and transition risks may affect downside outcomes differently from central outcomes.
- ▶ Climate and financial risks can interact and amplify each other.
- ▶ Slow-moving, fat-tailed and potentially irreversible dynamics make this an active research frontier.

- ▶ Quarterly US data: **1971Q1–2024Q3**.
- ▶ GDP growth: year-on-year real GDP growth.
- ▶ Inflation: year-on-year GDP-deflator inflation.
- ▶ NFCI: Chicago Fed National Financial Conditions Index, weekly data averaged to quarterly frequency.
- ▶ Positive NFCI values indicate tighter-than-average financial conditions.
- ▶ Forecast horizon: four quarters ahead.

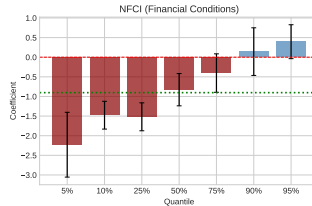
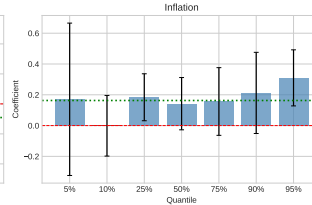
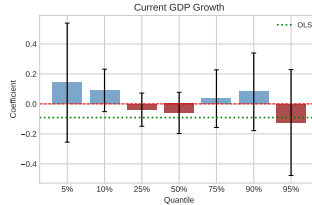
$$Q_{\tau}(\Delta y_{t+4} \mid \Omega_t) = \alpha(\tau) + \beta_y(\tau)\Delta y_t + \beta_{\pi}(\tau)\pi_t + \beta_f(\tau)NFCI_t$$

Estimated at $\tau \in \{0.05, 0.10, 0.25, 0.50, 0.75, 0.90, 0.95\}$.

Question

Does a tightening in financial conditions shift the entire growth distribution equally, or mainly worsen the downside tail?

Quantile Regression Coefficients
(Asymmetry = Different Effects on Downside vs Upside)



KEY FINDING: NFCI Asymmetry

NFCI effect on GDP growth:

- At 5th percentile: -2.229
- At 50th percentile: -0.827
- At 95th percentile: 0.395

Interpretation:

Tight financial conditions (NFCI > 0) have a LARGER negative effect on the left tail (downside risk) than on the median or right tail.

This is the key insight of the "Vulnerable Growth" framework: financial conditions predict the SHAPE of the growth distribution, not just its mean.

5th percentile

$$\hat{\beta}_f(0.05) = -2.23$$

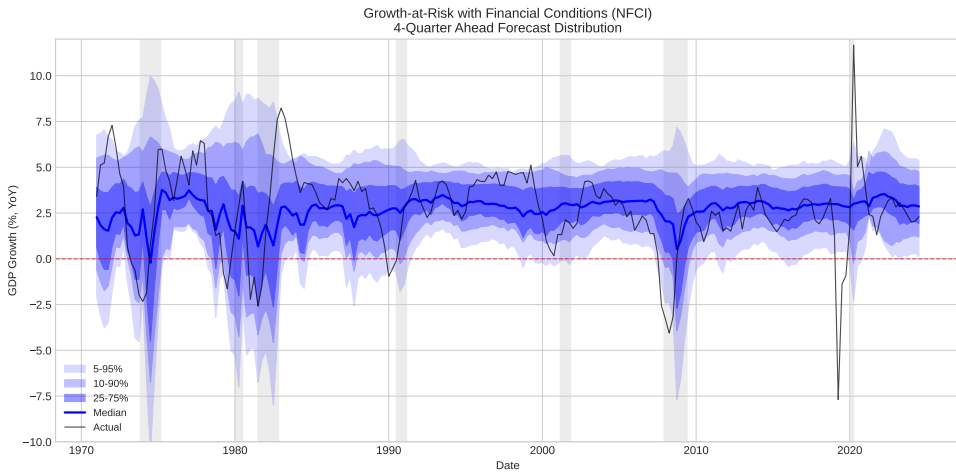
Median

$$\hat{\beta}_f(0.50) = -0.83$$

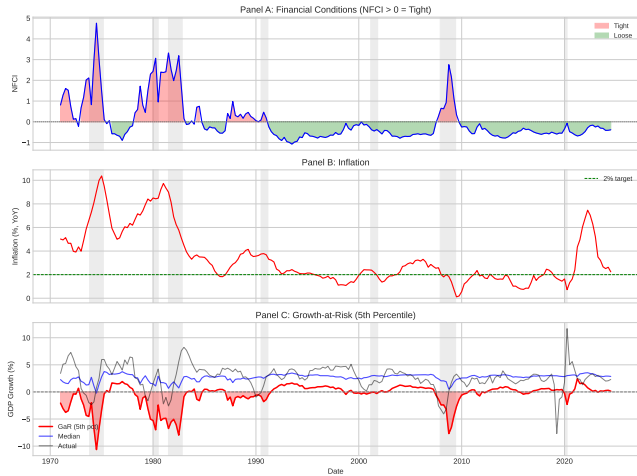
90th percentile

$$\hat{\beta}_f(0.90) = 0.14$$

- ▶ The downside effect is about **2.7 times larger** than the median effect.
- ▶ A one-standard-deviation NFCI tightening lowers the 5th percentile by 2.23 percentage points in the chapter estimates.
- ▶ The upper-tail effect is statistically insignificant.



- ▶ Predictive uncertainty changes over time rather than remaining constant.
- ▶ The distribution can become negatively skewed when financial conditions tighten.
- ▶ Lower percentiles tend to deteriorate before recession episodes.
- ▶ The Global Financial Crisis and the pandemic are associated with a dramatic widening of downside risk in the chapter application.



- ▶ Positive NFCI values correspond to tighter-than-average financial conditions.
- ▶ Tight conditions are associated with more negative GaR values.
- ▶ The relationship is especially visible around the early 1980s, 2008 and 2020 episodes highlighted in the chapter.
- ▶ The median forecast alone does not convey the same downside-risk information.

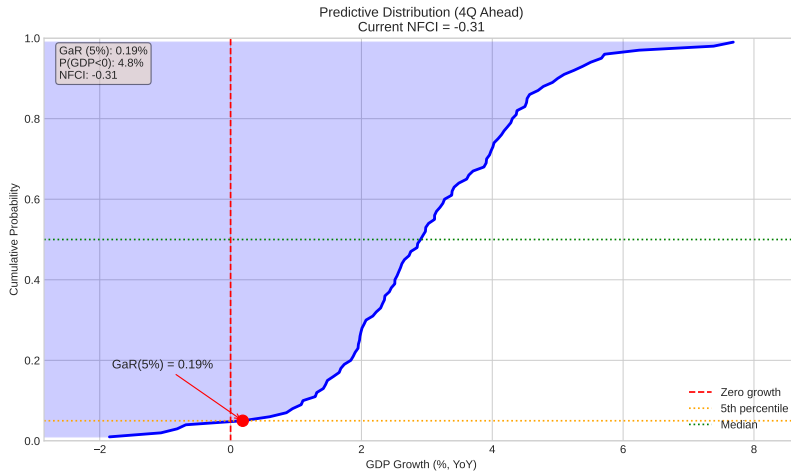
Quantile	Nominal	Empirical
0.05	5.0%	5.6%
0.10	10.0%	10.7%
0.25	25.0%	26.0%
0.50	50.0%	49.8%
0.75	75.0%	74.4%
0.90	90.0%	88.8%
0.95	95.0%	94.4%

Empirical coverage is close to nominal coverage across quantiles in the chapter application.

As of 2024Q3 in the empirical example

With NFCI around -0.31 , the fitted model reports GaR (5th percentile) of 0.19%, probability of negative growth of 4.8%, and a median forecast of 2.6%.

- ▶ These values describe the chapter's sample endpoint; they are not a current 2026 assessment.
- ▶ A hypothetical NFCI stress value of 2 shifts the fitted GaR to about -4.3% in the exercise.



1. Define the tail quantity and forecast horizon of interest.
2. Choose predictors motivated by the economic risk channel.
3. Estimate quantile-specific models over a suitable grid of τ .
4. Check coefficient heterogeneity and uncertainty across quantiles.
5. Enforce monotonicity if quantile crossing occurs.
6. Construct the predictive distribution and tail-risk measure.
7. Evaluate coverage and proper scores, ideally in real time.
8. Stress-test results across specifications, vintages and nonlinear extensions.

1. Quantile regression reveals distributional asymmetries hidden by conditional-mean models.
2. Financial conditions can matter far more for downside growth than for median or upside growth.
3. Growth-at-Risk summarizes the lower tail of the conditional growth distribution at a chosen horizon.
4. Quantile crossing, tail-sample size and real-time data require careful treatment.
5. The same framework extends naturally to inflation, systemic risk and climate-related tail risks.

- ▶ Estimate the GDP-growth equation at additional quantiles and plot $\hat{\beta}_f(\tau)$.
- ▶ Compare quantile-regression coefficients with OLS and explain the economic difference.
- ▶ Deliberately create quantile crossing and apply rearrangement.
- ▶ Compare one-quarter and four-quarter-ahead GaR forecasts.
- ▶ Evaluate tail forecasts using coverage and quantile scores.
- ▶ Replace NFCI with another financial-conditions or uncertainty indicator.
- ▶ Extend the specification to an Inflation-at-Risk or Climate-at-Risk exercise.

Companion slides = starting point

These slides summarize core concepts and selected examples from Chapter 12. They are intentionally editable and are not a substitute for the full chapter.

- ▶ Add your own risk indicators, countries and forecast horizons.
- ▶ Add coding demonstrations in Python, R, MATLAB or Stata.
- ▶ Add alternative density-reconstruction and scoring methods.
- ▶ Add empirical exercises on inflation, credit, financial stability or climate risk.

Make them yours.

Multi-Country Models

International linkages, spillovers and global dynamics

Next question

How can we model interconnected economies while allowing shocks to propagate across countries and markets?